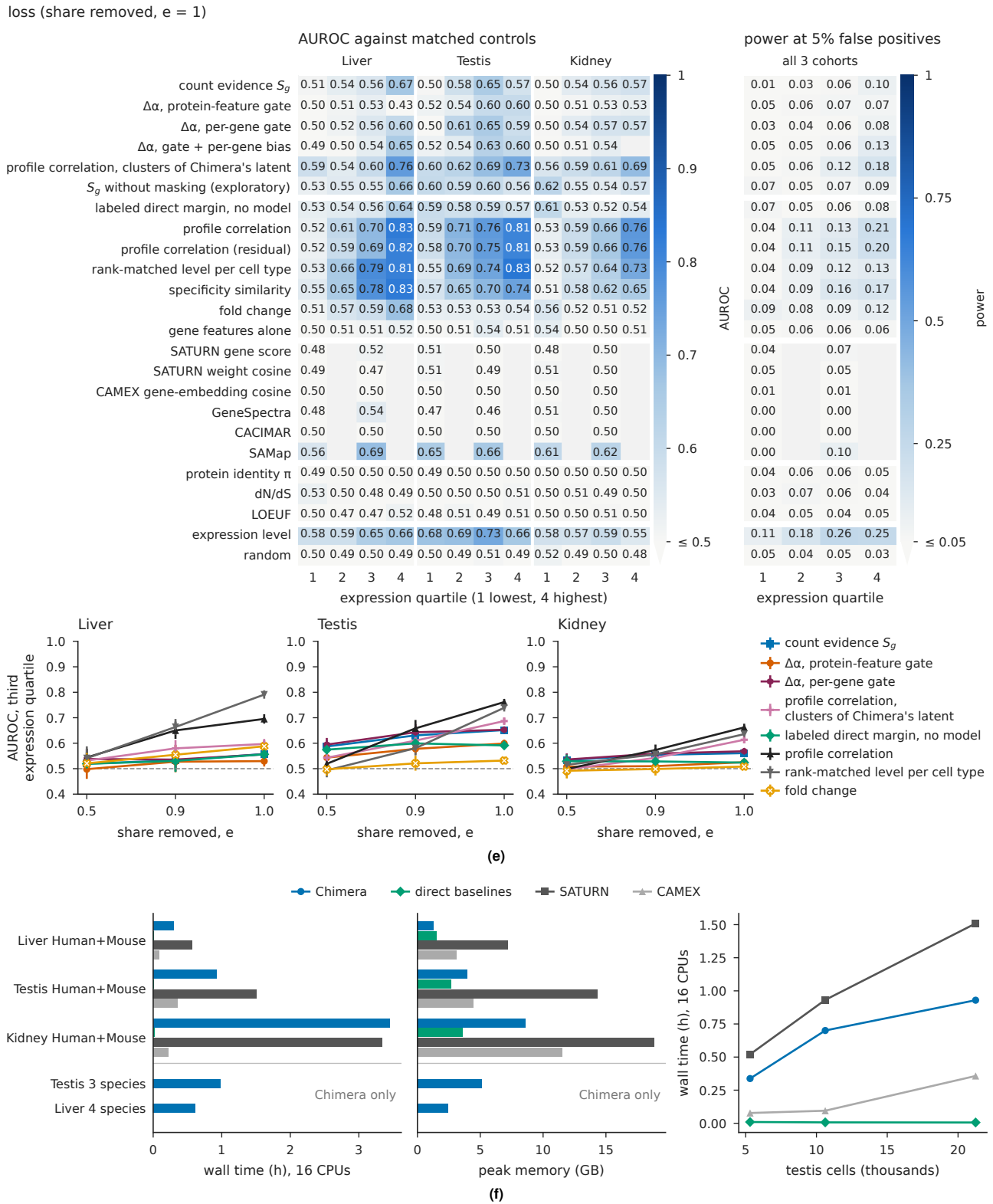

loss (share removed, e = 1)
AUROC against matched controls
power at 5% false positives
Liver
Testis
Kidney
all 3 cohorts
count evidence S_g
Δα, protein-feature gate
Δα, per-gene gate
Δα, gate + per-gene bias
profile correlation, clusters of Chimera's latent
S_g without masking (exploratory)
labeled direct margin, no model
profile correlation
profile correlation (residual)
rank-matched level per cell type
specificity similarity
fold change
gene features alone
SATURN gene score
SATURN weight cosine
CAMEX gene-embedding cosine
GeneSpectra
CACIMAR
SAMap
protein identity π
dN/dS
LOEUF
expression level
random
AUROC
power
≤ 0.5
≤ 0.05
expression quartile (1 lowest, 4 highest)
expression quartile
Liver
Testis
Kidney
AUROC, third expression quartile
share removed, e
count evidence S_g
Δα, protein-feature gate
Δα, per-gene gate
profile correlation, clusters of Chimera's latent
labeled direct margin, no model
profile correlation
rank-matched level per cell type
fold change
(e)
Chimera
direct baselines
SATURN
CAMEX
Liver Human+Mouse
Testis Human+Mouse
Kidney Human+Mouse
Testis 3 species
Liver 4 species
Chimera only
Chimera only
wall time (h), 16 CPUs
peak memory (GB)
wall time (h), 16 CPUs
testis cells (thousands)
(f)